

Steady State Cost in Competing Aggregation-Fragmentation Dynamics

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ABSTRACT

Aggregation and fragmentation among polymers has long been an area of active research and utility in physical chemistry. Drawing inspiration from the dynamics of social formulation, we make a preliminary effort at minimizing cost and probability of domination among various potential fragmentation kernels. This work has relevance for any work formalizing the formulation, dissolving, and regulation of groups of abstract individuals. Social groups are of particular interest, from musical subcultures to the regulation of corporate consolidation.

1 Introduction

Things fall apart, Yeats wrote¹. Of course, they also come together. What balance is struck between the two, and crucially, **what does this balance cost?**

To attempt an answer to this question, we turn to the physics of aggregation and fragmentation, the competing processes by which particles combine to form larger clusters and break up into smaller ones. As thoroughly explicated in *A Kinetic View of Statistical Physics*, aggregation and fragmentation can be thought of as an infinite set of differential equations over all cluster sizes from one to infinity². Rates of change are based on current amounts of clusters and a kernel specifying reaction rates between clusters of various sizes. Clusters of size one are termed monomers, and the formulation as a whole is termed the Smoluchowski equations.

While Krapivsky *et al* term such differential equations master equations, their equations make a mean-field assumption: they track the *expected* cluster size distribution and assume it deterministic rather than keeping track of the full spread of probability over the stochastic evolution of the system. In other words, they track the *count* of the cluster sizes, rather than the probability of *each count* of each cluster.

A more complete accounting is given by the Marcus–Lushnikov process, in which $ML^{(n)}(x, t)$ is a random number whose probability evolves according to a Markov Chain involving all other $ML^{(n_i)}(x, t)$ ³. These models typically rescale the total mass of the system to be 1, and are identical to the Smoluchowski equations if the kernel is homogenous, or as N , the number of initial unitary particles considered, goes to infinity. However, these models are far less tractable, and even in 2018 computing the time-evolving probability with only 48 initial monomers took around 3000 minutes⁴.

Regardless of the form, prior work has largely focused on either the time-evolution of cluster sizes or the static steady state distribution of them. Instead, this work focus on the interplay between the steady state rate of fragmentation, or the *cost* of maintaining the steady state, and the distribution of cluster sizes. More simply, using the Smoluchowski equation form, this work charts the pareto front between minimizing the steady state probability of having an oversized *dominant* cluster and minimizing the number of units of mass continually split by fragmentation.

2 Methods:

The problem

We model competing aggregation and fragmentation over a variety of rate kernels in order to study the steady-state distribution of cluster sizes as well as the steady-state rate of fragmentation measured in units of mass removed from larger clusters. Our question is simple: given a pattern of aggregation, what fragmentation technique minimize the proportion of dominant clusters and steady-state fragmentation rate? Does this dual-minimization create a pareto front?

The model

Aggregation

Aggregation is generally the transition

$$c_i + c_j \xrightarrow{K_{ij}} c_{i+j}$$

where:

- N is the total amount of mass divided by the size of the smallest cluster, which we normalize to 1.
- c is a cluster
- i and j are masses
- K_{ij} is the rate of aggregation between clusters of mass i and mass j into a clusters of mass $i + j$
- K , then, is an finite symmetric matrix $K_{i,j} = K_{j,i}$ of dimensions $N \times N$

With the ordinary differential equation for all cluster sizes c_k :

$$\frac{dc_k}{dt} = \dot{c}_k = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{i \geq 1} K_{ik} c_i \quad (1)$$

$$= \frac{1}{2} \sum_{i,j \leq 1} K_{ij} c_i c_j \left[\delta_{i+j,k} - \delta_{i,k} - \delta_{j,k} \right] \quad (2)$$

where δ is the Kronecker delta.

Fragmentation

Fragmentation is generally the transition

$$c_{i+j} \xrightarrow{F_{ij}} c_i + c_j$$

where F is an finite symmetric matrix of size $N \times N$ and F_{ij} is the rate of fragmentation of a cluster of size $i + j$ into a cluster of size i and another of size j .

The ordinary differential equation for fragmentation across all clusters is then::

$$\frac{dc_k}{dt} = \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} \sum_{i+j=k} F_{ij} \quad (3)$$

Aggregation and Fragmentation

Putting equations (1) and (3) together yields the overall differential equation:

$$\frac{dc_k}{dt} = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{j \geq 1} K_{kj} c_j + \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} c_k \sum_{i+j=k} F_{ij} \quad (4)$$

Cost and Domination

Kernels

Table 1

Third-level section

3 Results

The results can contain subheadings as desired and can discuss the results directly.

4 Discussion

The Discussion should be succinct. It typically involve any big picture discussions not covered in Sec. 3, i.e., the Results section. This was an example of a section reference using the `label` and `ref` commands. Figures and tables can also be referenced in LaTeX using the `label` and `ref` commands, e.g. Fig. 1 and Tab. 3.

This discussion can therefore mention scientific consequences of the results, ideas for future work, stress approximations or simplifications that limit the applicability of the model, or discuss things you tried that did not work. The Discussion section must not contain subheadings.

Name	Equation
Chipping Model	$F_{ij} = \gamma(\delta_{i,1} + \delta_{j,1})$
Chipping Model	$F_{ij} = \gamma(\delta_{i,5} + \delta_{j,5})$
Sum	$F_{ij} = \gamma(i + j)$
Product	$F_{ij} = \gamma(ij)$
Constant	$F_{ij} = \gamma$
Reverse Product	$F_{ij} = \frac{\gamma}{ij}$
Dunbar	$F_{ij} = \gamma\left(ij + \frac{2(i+j)}{150} + \frac{4}{150^2}\right)$

Table 1. Specification of the seven kernels used for aggregation and fragmentation.

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Additional information

Data availability statement: No real world data is used in this project

Code availability statement: Code is available at github.com.

References

1. Yeats, W. *The Second Coming*.
2. Krapivsky, P. L., Redner, S. & Ben-Naim, E. *A Kinetic View of Statistical Physics* (Cambridge University Press, 2010), 1 edn.
3. Aldous, D. J. Deterministic and stochastic models for coalescence (aggregation and coagulation): A review of the mean-field theory for probabilists. *Bernoulli* **5**, 3–48 (1999).
4. Manuchehrfar, F., Tian, W., Chou, T. & Liang, J. Evolution of Coagulation-Fragmentation Stochastic Processes Using Accurate Chemical Master Equation Approach. *Commun. Inf. Syst.* **19**, 37–55, DOI: [10.4310/cis.2019.v19.n1.a3](https://doi.org/10.4310/cis.2019.v19.n1.a3) (2019).



Figure 1. Legend (350 words max). Example legend text for an example figure. Figures should be in pdf format unless it is a photo or archival image not in vector graphics, as is the case here (in which case jpg and png are acceptable).

Table 2. Model Card

1. Model Details		
The problem		
Modeling approach		
Degrees of freedom		
Number of parameters		
Implementation language		
Relevant publications		
License		
Contact information		
2. Intended Use		
Primary intended uses		
If forecasting, validation metrics		
If applicable, data incorporated		
Primary users		
Out-of-scope uses		
3. Key assumptions		
Population size		
Contact structure		
Time structure and memory effects		
Other		
4. Metrics		
Computational complexity		
Variation approaches		
8. Ethical Considerations		
Potential risks and harms		
Bias and fairness concerns		
Mitigation strategies		
Impact on stakeholders		
9. Suggested next steps		
Issues with implementations		
Recommendations for use		
Future work		

Condition	n	p
A	5	0.1
B	10	0.01

Table 3. Legend (350 words max). Example legend text for an example table.